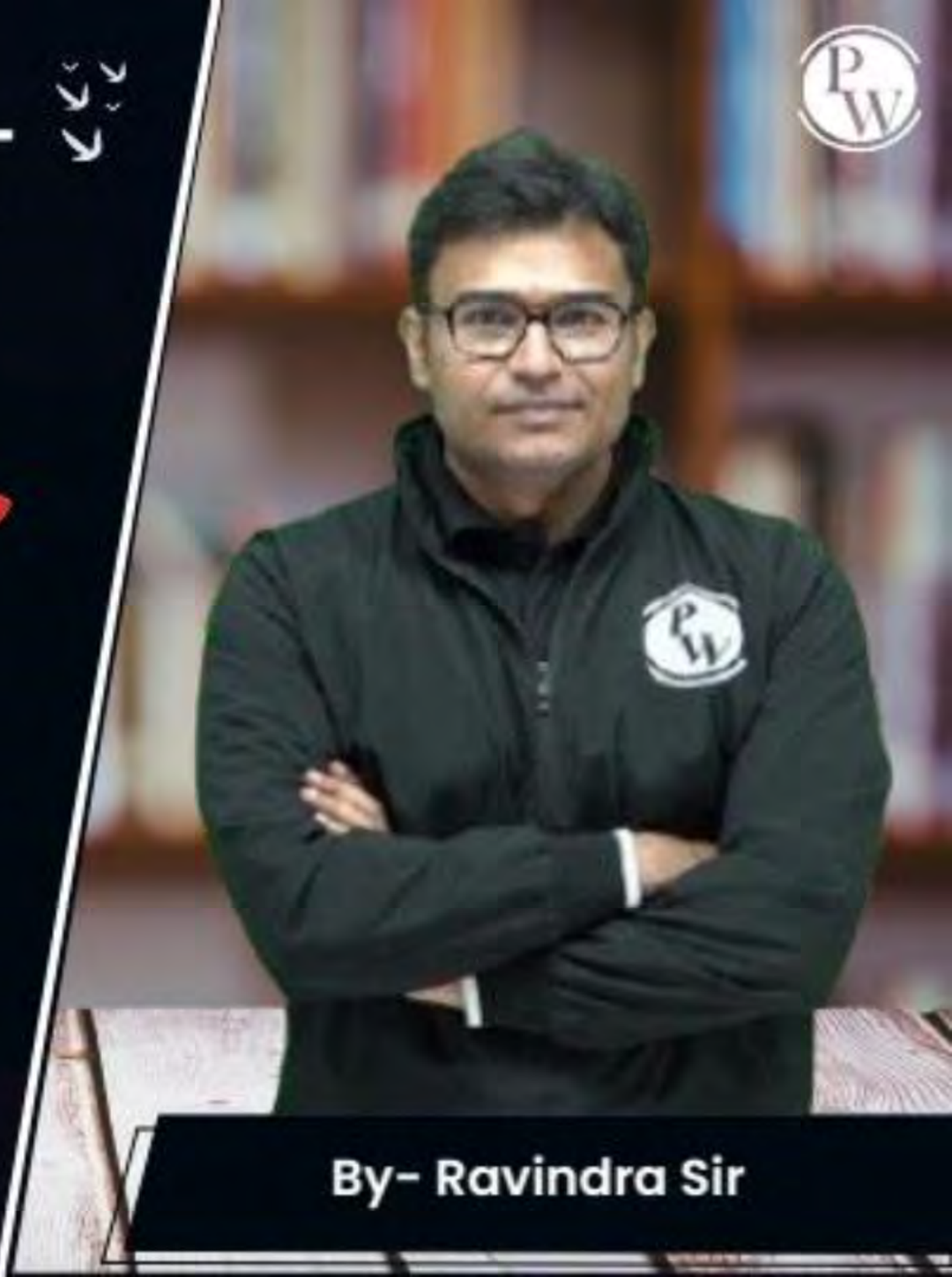
Computer Science & IT

Theory of Computation



Regular Languages

Lecture No. 3



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

DFA, NFA, RE

Topics to be Covered



Topic

Topic

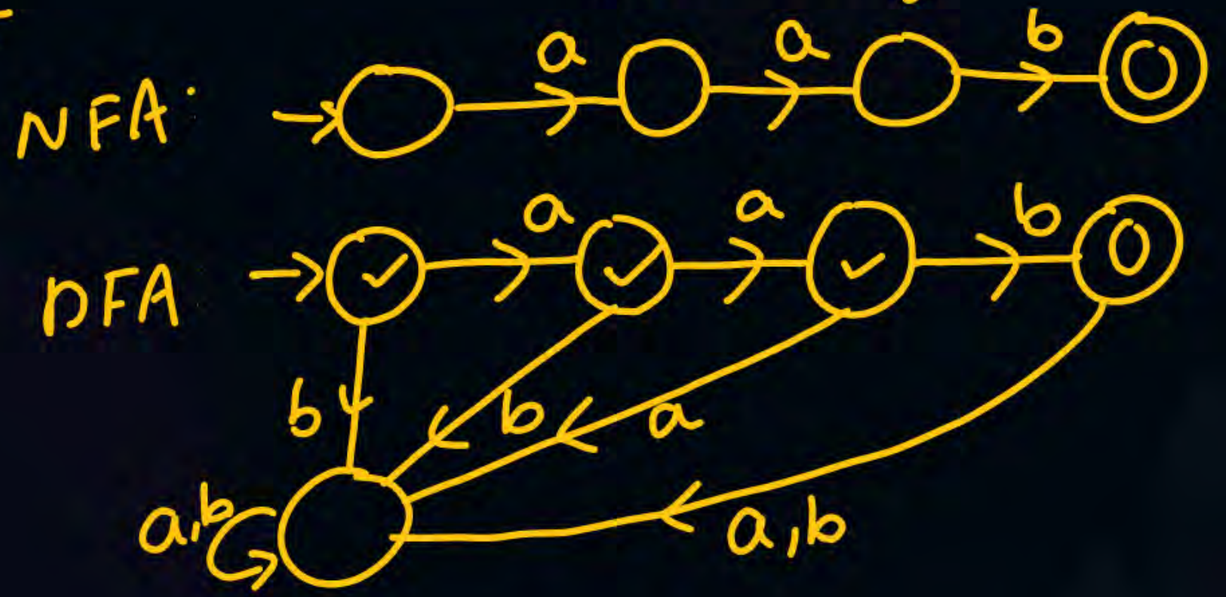
Topic

NFA, DFA

$\Sigma = \{a, b\}$ $L = \{aab\}$ RE = aab

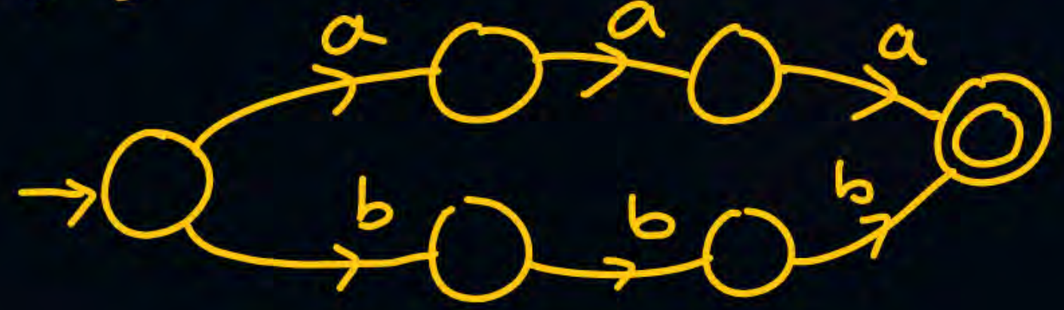
$aab \neq a^2b$

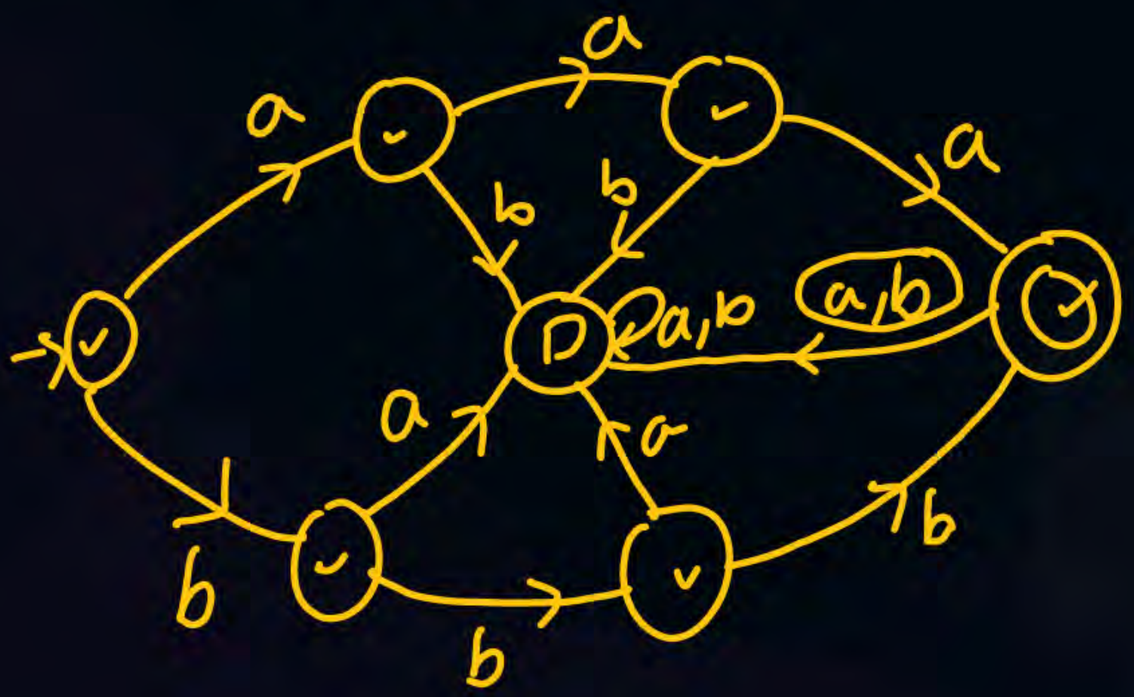
aab → one string



$\Sigma = \{a, b\}$ $L = \{aaa, bbb\}$ $RE = aaa + bbb$

NFA:





$\Sigma = \{a, b\}$ $L = \{w / |w| \text{ is even}\}$ $RE = ((a+b)(a+b))^*$

$L = \{\epsilon, aa, ab, ba, bb, aaaa, aaab, \ldots\ldots\}$

$(aa + ab + ba + bb)^* = (a(a+b) + b(a+b))^*$

$((a+b)(a+b))^*$



min DFA
min NFA

$A \xrightarrow{a} B$

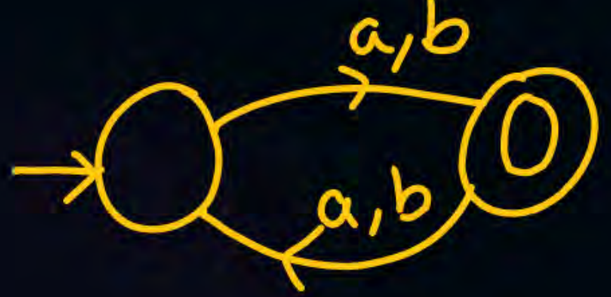
$A \xrightarrow{b} B$

$B \xrightarrow{a} A$ $B \xrightarrow{b} A$



$\Sigma = \{a, b\}$ $L = \{w / |w| \text{ is odd}\}$

$2n+1$ $\underbrace{\underbrace{((a+b)(a+b))^*}_{\text{even}}(a+b)}$



$(a+b)((a+b)(a+b))^*$

$(a(a+b)+b(a+b))^*$

$((a+b)(a+b))^*$

min DFA, min NFA

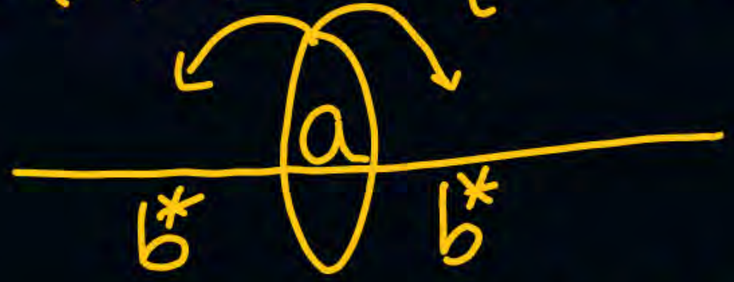
$\Sigma = \{a, b\}$ $L = \{ w / |w| \text{ is } \cong 2 \bmod 3 \}$ divisible by (n)

$(a+b)(a+b)(a+b)(a+b)(a+b)(a+b)^*$ → mod machine

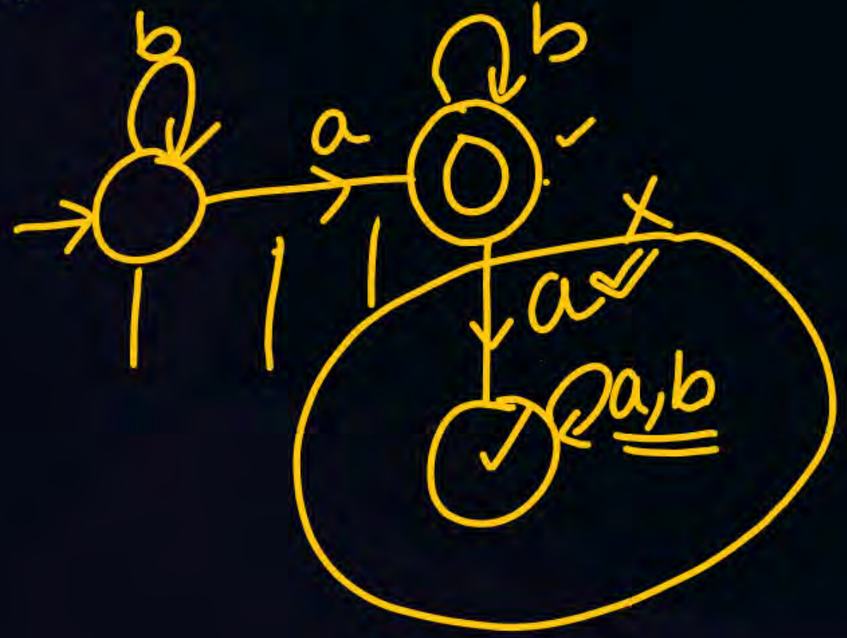
$((a+b)(a+b)(a+b))^*(a+b)(a+b)$ $3n+2$

$(a+b)((a+b)(a+b)(a+b))^*(a+b)$

$\Sigma = \{a, b\}$ $L = \{ w / w \text{ has exactly 1 } a \}$



$L = \{ a, ba, ab, bba, bab \ldots \}$



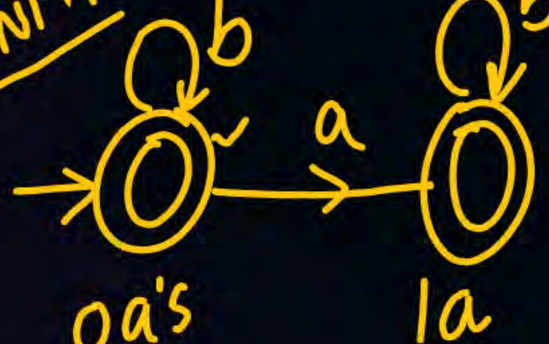
$\Sigma = \{a, b\}$ $L = \{w / w \text{ has atmost 1 } a\}$

Atmost 1a = 0 a + 1a

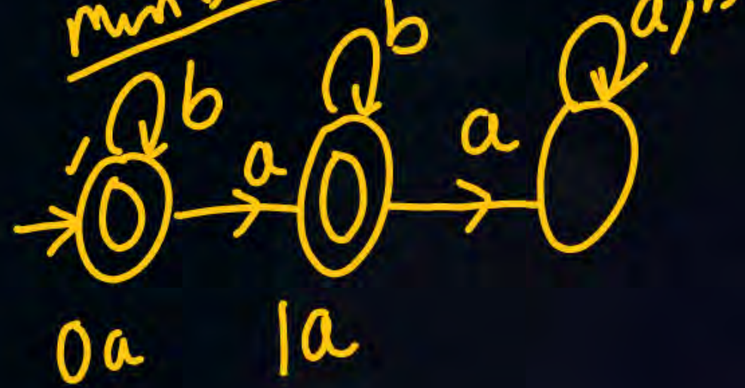
$b^* + b^*ab^*$ $+ b^* + b^*ab^*$

$L = \{\epsilon, b, bb, bbb, \cdots, a, ab, ba \cdots ---\}$

min NFA

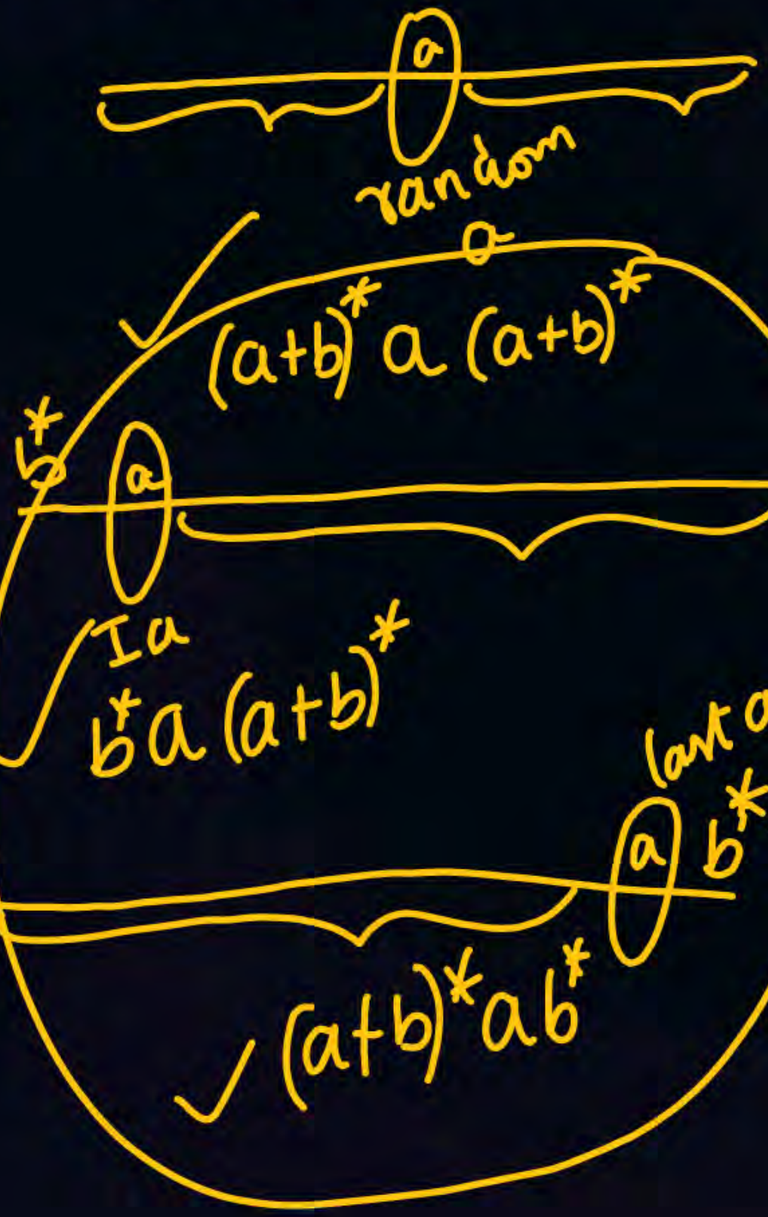


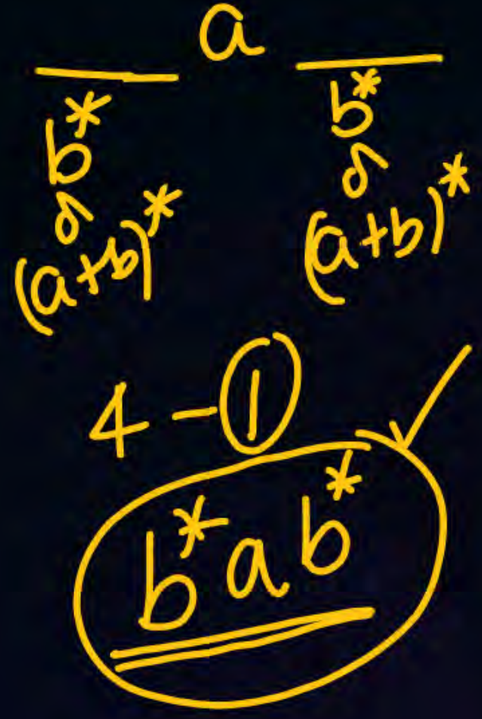
min DFA



$\Sigma = \{a, b\}$ → $L = \{ w \mid w \text{ contains atleast 1a} \}$

$L = \{a, ab, ba, aa, aab, \ldots\}$





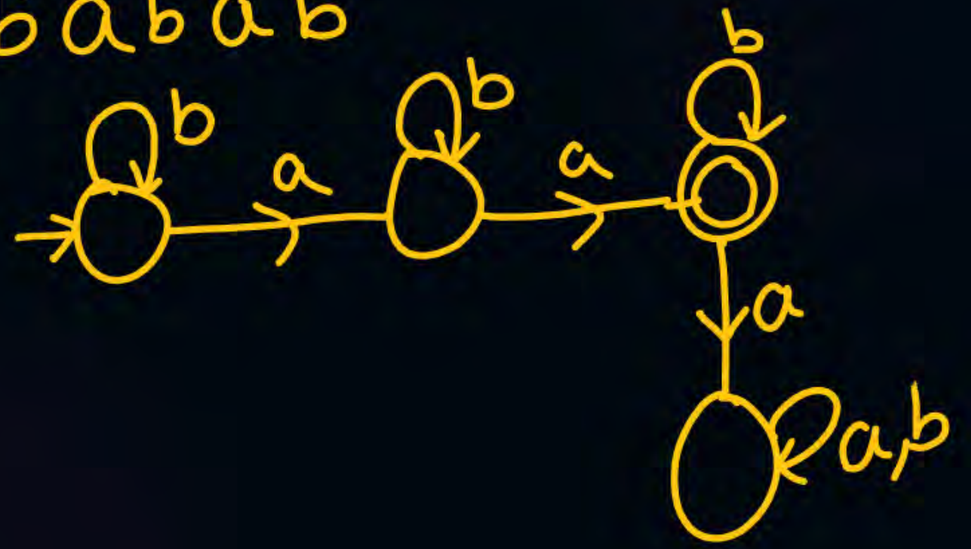


min DFA
min NFA

$\Sigma = \{a, b\}$ $L = \{w / w \text{ has exactly 2 a's}\}$

$b^* a b^* a b^*$

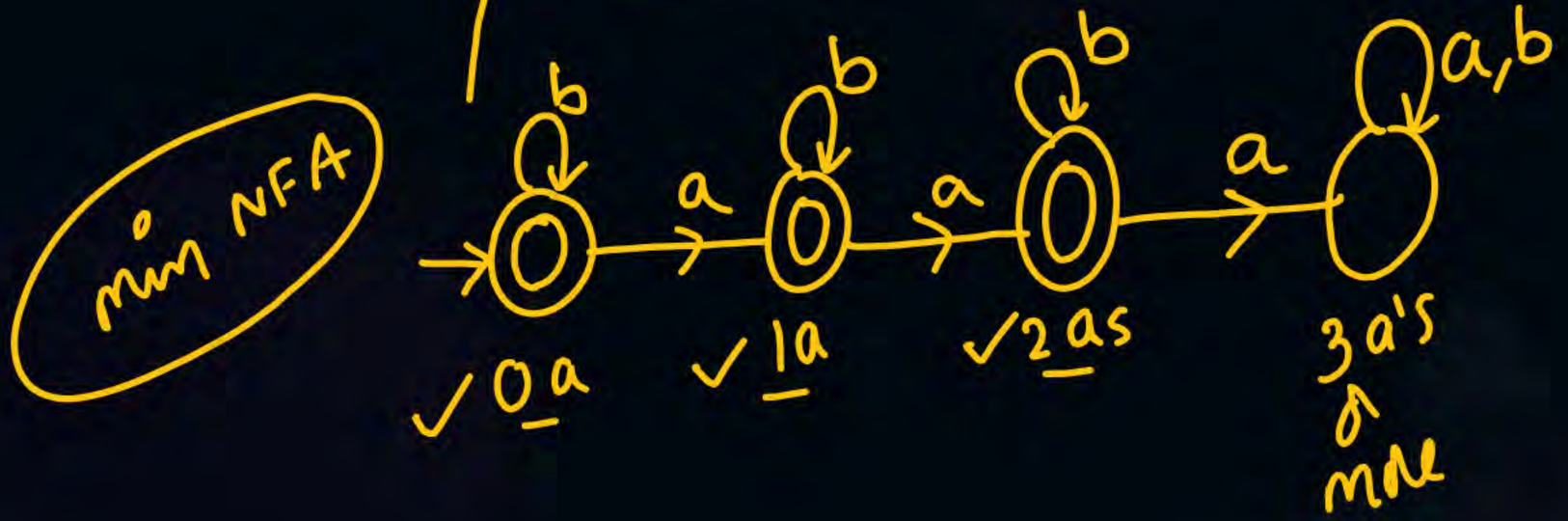
min NFA



$\Sigma = \{a, b\}$ $L = \{w / w$ contains atmost 2 a's$\}$

0a 1a 2a

$\epsilon + b^* + b^*ab^* + b^*ab^*ab^*$



THANK - YOU